

Seminar Report

Polynomials and High-Dimensional Spheres

(talk by Prof. Ambar Sengupta)

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Introduction

This report is on the talk *Polynomials and High-Dimensional Spheres* given by Prof. Ambar Sengupta of the University of Connecticut on November 12, 2024.

The talk opened with a brief historical primer and was organised into four parts: Gaussian integration and spherical integration; harmonic polynomials and operators on polynomial spaces; Hermite limits over large spheres; and the spherical Laplacian in higher dimensions.

Three key results were highlighted. First, the spherical Laplacian converges to the Hermite differential operator:

$$\lim_{N \rightarrow \infty} \Delta_{S^{N-1}(\sqrt{N})} = \sum_{i=1}^{\infty} (\partial_i^2 - X_i \partial_i).$$

Second, Gegenbauer polynomials converge to Hermite polynomials:

$$\lim_{N \rightarrow \infty} q_m(\sqrt{N}; X) = \lim_{N \rightarrow \infty} C_m^{(N-2)/2}(X/\sqrt{N}) = H_m(X),$$

where $q_m(\sqrt{N}; X)$ coincides with a zonal harmonic polynomial on the sphere $S^{N-1}(\sqrt{N})$. Third, Gaussian integration arises as the limit of spherical integration:

$$\langle p, q \rangle_{L^2(\mathbb{R}^\infty, \mu)} = \lim_{N \rightarrow \infty} \langle p, q \rangle_{L^2(S^{N-1}(\sqrt{N}), \bar{\sigma})},$$

where $p, q \in \mathcal{P}_k$ are viewed in $\mathbb{R}^N$.

This third result is remarkable: it simplifies the infinite-dimensional spherical Laplacian — which is already cumbersome in finite dimensions — to a tractable and concise Hermite differential operator. The use of rotation generators, the quadratic Casimir, and the restriction–projection map onto the sphere (which produces a unique harmonic polynomial) was particularly intuitive. One motivation mentioned in the talk lies in statistical mechanics, where the total kinetic energy of a system can be written as a function of velocities indexed by particles. As the number of particles grows, the total kinetic energy becomes increasingly difficult to handle directly, and the results of this talk offer a way to simplify these kinds of problems.

The report below summarises each part of the talk. It draws on a careful reading of the accompanying paper by Amy Peterson and Ambar Sengupta of the University of Connecticut.

Gaussian Integration as the Limit of Spherical Integration

The plan of attack in this part was first to establish a relation between the spherical integral of a homogeneous (Borel-measurable) polynomial function and the corresponding Gaussian integral. The relation is intuitive: the Gaussian integral over $\mathbb{R}^N$ can be written as a coefficient times the integral of the same function over the unit sphere. Explicitly, the coefficient is

$$A_{d,N} = 2^d \frac{\Gamma\left(\frac{d+N}{2}\right)}{\Gamma\left(\frac{N}{2}\right)},$$

where d is the degree of the function. The L^2 pairing on the sphere is then related to the Gaussian L^2 pairing by

$$a_{d,N} \langle p, q \rangle_{L^2(S^{N-1}, \bar{\sigma})} = \langle p, q \rangle_{L^2(\mathbb{R}^N, \mu)},$$

where

$$a_{d,N} = \prod_{j=1}^d \left(1 + \frac{2j-1}{N}\right) \quad \text{and} \quad d = \frac{d_p + d_q}{2}.$$

This can be shown straightforwardly by polar disintegration and simplification via the Gamma function.

The interesting consequence is that for $p, q \in \mathcal{P}_k$ with $k < N$, the spherical L^2 pairing forms an inner product on $\mathcal{P}_k$. Since $\mathcal{P}_k = \bigoplus_{d=0}^{\infty} \mathcal{P}^d$ and $a_{d,N} \xrightarrow{N \rightarrow \infty} 1$, we obtain

$$\lim_{N \rightarrow \infty} \langle p, q \rangle_{L^2(S^{N-1}, \bar{\sigma})} = \langle p, q \rangle_{L^2(\mathbb{R}^{\infty}, \mu)}.$$

Harmonic Polynomials and Operators on Polynomial Spaces

This part was dedicated to developing the relevant operators and their actions on polynomial spaces.

If we endow the polynomial space $\mathcal{P}$ with a Hermite inner product, then the adjoint of ∂_j is simply multiplication by X_j , where X_j is the corresponding variable. It follows easily that the adjoint of Δ_N is multiplication by $\|X\|^2 = X_1^2 + \dots + X_N^2$.

Since $\Delta_N : \mathcal{P}_N^d \rightarrow \mathcal{P}_N^{d-2}$ is a linear operator, we can split

$$\mathcal{P}_N^d = \ker \Delta_N \oplus \|X\|^2 \mathcal{P}_N^{d-2} = \mathcal{H}_N^d \oplus \|X\|^2 \mathcal{P}_N^{d-2}.$$

This extends naturally to $\mathcal{P}_N^{\leq d}$. Thus, for any $p \in \mathcal{P}_N$, we have $p = p_0 + \|X\|^2 q$ with $p_0 \in \mathcal{H}_N$ and $q \in \mathcal{P}_N$. More generally, any $p \in \mathcal{P}_N^{\leq d}$ can be expanded as

$$p = p_0 + \|X\|^2 p_1 + \dots + \|X\|^{2s} p_s,$$

where $p_j \in \mathcal{H}_N^{\leq d-2j}$.

The operators L_a (which generates a unique harmonic polynomial), M_{jk} and $\|M\|^2$, together with their respective algebraic properties, were then developed. These tools were used to derive zonal spherical harmonics. By definition, a zonal harmonic is a harmonic polynomial that is invariant under all rotations fixing one vector. To simplify the algebra, we may assume $t = (1, 0, 0, \dots, 0)$, since any vector can be obtained from t by scaling and rotation. If $p \in \mathcal{P}_N$ is invariant under all rotations preserving a given direction, then $p = q(a; X)$ for all $x \in S^{N-1}(a)$. On the sphere

$L_a p = L_a q(a; X)$, and $L_a q(a; X) \in \mathcal{H}_N$ is homogeneous. Because this polynomial is invariant under rotations preserving a direction, harmonic, and homogeneous, it is a zonal harmonic. This is how zonal harmonics may be generated, with the distinction that when restricted to the sphere $q(a; X)$ is itself a zonal harmonic. Concretely,

$$C_m^{(N-2)/2}(X/a) = q_m(a; X)$$

when restricted via L_a . This identity is used later in conjunction with orthogonality under $\langle \cdot, \cdot \rangle_{S^{N-1}(a)}$ to obtain the Hermite limit.

Since $\langle \cdot, \cdot \rangle_{a,N}$ is an inner product on $\mathcal{P}_k$ when $k < N$, the adjoints of the operators in use can be found by appealing to the earlier relation between spherical and Gaussian integration. The properties of these operators are then used to show that $\mathcal{H}_N^m$ is orthogonal to $\mathcal{H}_N^n$ when $m \neq n$. From this orthogonality of the zonal harmonics, one derives $q_m(a; X_1)$, monic with $q_0(X_1) = 1$, satisfying

$$q_m(a; X_1) = (I - P_{a,N}) X_1^m,$$

where $P_{a,N}$ is the projection from $\mathcal{P}_N^{\leq m}$ onto $\mathcal{P}_N^{\leq m-1}$ with respect to $\langle \cdot, \cdot \rangle_{a,N}$. Furthermore, applying Gram-Schmidt to $1, X, X^2, \dots$ under this inner product produces the zonal harmonics directly.

Hermite Limits

In this part the Gaussian L^2 inner product was revisited. The projection $P_N'^{\leq d} : \mathcal{P}_N^{\leq d} \rightarrow \mathcal{P}_N^{\leq d-1}$ extends to all of $\mathcal{P}^{\leq d}$, since the restriction of the projection in M variables to $\mathcal{P}_N^{\leq d}$ agrees with the projection in N variables.

Using this, Hermite polynomials are produced from monomials:

$$(1 - P'^{\leq d})(X_1^{m_1} \dots X_N^{m_N}) = H_{m_1}(X_1) \dots H_{m_N}(X_N),$$

with $m_1 + \dots + m_N = d$.

Using the fact that if a sequence of inner products $\langle \cdot, \cdot \rangle_n \rightarrow \langle \cdot, \cdot \rangle$, then the orthogonal projections P_n onto a fixed proper subspace converge to the projection P under the limiting inner product, and combining this with the limiting relation between the Gaussian and spherical inner products from the first part, we obtain

$$\lim_{N \rightarrow \infty} (1 - P_{k,N}^{\leq d})(X_1^{m_1} \dots X_k^{m_k}) = \lim_{N \rightarrow \infty} (1 - P'^{\leq d})(X_1^{m_1} \dots X_k^{m_k}) = H_{m_1}(X_1) \dots H_{m_k}(X_k).$$

The limiting relation stated in the introduction between zonal harmonics, Gegenbauer polynomials, and Hermite polynomials then follows.

Spherical Laplacian

In this part the properties of the spherical Laplacian were discussed first. The operator $\Delta_{N-1,a}$ acts on $p \in \mathbb{C}[X_1, \dots, X_n]$ in two cases. First, when $n \leq N$,

$$\Delta_{N-1,a} = \frac{1}{a^2} \|M\|^2,$$

and second, when $n > N$,

$$\Delta_{N-1,a} p = \frac{1}{a^2} \|M\|^2 p = q_{\min},$$

where $q_{\min} \in \mathbb{C}[X_1, \dots, X_n]$ is the unique minimal representative modulo the ideal $\mathcal{Z}_N(a)$ of polynomial multiples of $(\|X\|^2 - a^2)$. This is demonstrated by exploiting the self-adjointness of the spherical Laplacian with respect to $\langle \cdot, \cdot \rangle_{a,N}$, together with linearity. Examining the action on monomials in the limit $N \rightarrow \infty$, one finds that

$$\Delta_{N-1, \sqrt{N}} \longrightarrow \sum_{i=1}^{\infty} (\partial_i^2 - X_i \partial_i)$$

for any polynomial in $\mathcal{P}_N^{\leq m}$.

Remarks

Both the talk and the accompanying paper were straightforward to follow and genuinely intuitive, thanks to the systematic use of operators on polynomial spaces.

References

Peterson, A., & Sengupta, A. N. (2019). *Polynomials and High-Dimensional Spheres*. [arXiv:1903.07697](https://arxiv.org/abs/1903.07697) [[math.MP](#)].